Harinath Reddy Anugula

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Experience

Deloitte USI |***Data and AI Eningeer II*** | ***AWS, FastAPI, Docker, MCP, AI agents, RAG*** **Sep 2024 – Present**

- Led development of enterprise-grade Generative AI systems using Azure OpenAI, Azure ML, and Cognitive Services, designing scalable RAG pipelines and agentic AI architectures with LangChain/LangGraph for intelligent knowledge retrieval and automation
- Built and optimized vector embedding pipelines and semantic retrieval systems using ChromaDB, improving response accuracy and reducing hallucinations in production environments
- **Underwriter Ally – AI Copilot for Bond and Surety Insurance**
  - Built a Graph RAG copilot that helps underwriters draft coverage and answer submission questions across about 300 documents (strategy guides, quick references, exclusions), piloted with up to 50 underwriters.
  - Evaluated with LLM-as-judge against a golden set of about 100 conversations, benchmarked against the earlier plain-RAG chatbot and validated with underwriter feedback.
  - Designed and deployed a multi-agent AI system with memory, planning, and tool orchestration using LangGraph/MCP and Neo4j knowledge graphs for complex reasoning over policies.
  - Automated complex underwriting queries (risk appetite, eligibility, coverage terms) with 85–90% accuracy, enhancing decision consistency.
- **AI Based Clause Review Engine — Temporal, AWS Textract, TF-IDF, LLMs, EKS, Helm, ArgoCD, GitHub Actions, Terraform, Kibana, Dynatrace**
  - Built and shipped a production document-review pipeline that processes 100-200 insurance slips per day (about 8 pages each) to find clauses and flag typos or missing words against the original wording..
  - Orchestrated a multi-stage workflow on Temporal: document processing, then clause identification on each page using 3 sequential LLM calls, then comparison with the original clause.
  - Built observability in Kibana and Dynatrace, with dashboards and alerts that catch LLM processing errors and failed calls.
  - Built CI/CD pipelines for seamless deployment.
- **A/B Testing Framework for AI Agents — Amazon Bedrock AgentCore Gateway, Terraform, AWS**
  - Built a gateway-based experimentation layer that lets teams A/B test prompt and agent changes in dev. A new prompt reaches only a chosen set of users or a percentage of requests, with no redeploy of the agent.
  - Managed Terraform scripts and DevOps for enterprise deployment.
  - Provisioned the AgentCore Gateway and routing configuration as infrastructure-as-code with Terraform, so experiments are reproducible and reviewable.
- **STTM – Informatica Source To Target Mapper**
  - Engineered an AI-powered system to automatically generate source-to-target data lineage from Informatica XML mappings using agentic architecture (MCP).
  - Reduced manual analysis effort by transforming complex ETL logic into structured, interpretable lineage steps.
  - Built scalable processing pipelines for handling large XML mapping files in enterprise environments.

Veltris Technologies | ***AI Engineer*** **Aug 2021 – Sep 2024**

Implemented end-to-end MLOps pipelines including CI/CD, Docker-based containerization, and scalable cloud deployment for AI systems

- **NLP-Based Invoice Evaluation System (Law Firm)**
  - Built and integrated an NLP-based system to evaluate and classify attorneys' daily work entries at Kaufman Dolowich.
  - Trained a custom Named Entity Recognition (NER) model to extract task-related entities from free-text descriptions.

* Designed a fuzzy matching system to distinguish between billable and non-billable activities.
* Improved accuracy of task identification and automated invoice evaluation workflows.

- **Risk Quantification System for Project Tenders (Hitachi Energy)**
  * Built and optimized large-scale data pipelines using PySpark and Databricks, orchestrated via Azure Data Factory, to transform raw data into gold-standard datasets for downstream ML and analytics.
  * Developed machine learning classification models (Random Forest, XGBoost, Decision Tree) for risk prediction and probabilistic modeling, achieving 80–88% accuracy in identifying high-risk tenders
  * Built an end-to-end ML pipeline including data preprocessing, feature engineering, model training, and evaluation, reducing manual risk analysis effort by 35%
- **AI Assistant for Videos** |***Multi-Modal LLM (LLaVA-13B), Hugging Face, AWS, ChromaDB, Flask, APIs***
  * Developed a multi-modal GenAI assistant for Veltris capable of processing text, images, and video inputs.
  * Implemented in-depth video analysis to generate concise and accurate summaries of video content.
  * Enabled interactive chat functionality allowing users to ask contextual questions about video content.
  * Designed and managed vector databases (ChromaDB) for efficient semantic search and retrieval in RAG systems.
  * Deployed the system on AWS with scalable API endpoints using Flask.
- **Safety Surveillance System on Industry Shop-floor** |***Ultralytics, OpenCV, YOLOv8***
  * Designed and developed a real-time computer vision system to monitor worker safety in a manufacturing environment for Juniper.
  * Trained a custom YOLOv8 object detection model to verify compliance with safety gear (helmets, gloves, jackets, etc.).
  * Built a facial recognition module to identify workers not wearing required safety equipment.
  * Integrated real-time video streams for continuous detection and alerting of safety violations.
  * Improved on-site safety monitoring by automating visual inspection processes.
- **LLM for Customer Support** |***Hugging Face, A100 GPU, CosmosDB, AWS, FastAPI***
  * Fine-tuned multiple variants of the FALCON LLM (7B, 13B, 40B) for customer support use cases in the electronics domain.
  * Built a Retrieval-Augmented Generation (RAG) pipeline to improve response relevance and factual accuracy.
  * Designed APIs using FastAPI to serve the model for real-time customer query resolution.
  * Stored and retrieved domain knowledge using CosmosDB to support contextual answer generation.
  * Optimized model training and inference on NVIDIA A100 GPUs for high-throughput workloads.
- **Video Processing Pipeline** |***Airflow, Docker Compose, DAGs, AWS S3, AWS Redshift***
  * Engineered an Apache Airflow pipeline to automate a video anonymization workflow for Hexagon.
  * Built frame-level processing by segmenting videos into images for facial detection and blurring.
  * Implemented face recognition to identify and anonymize sensitive visual data in video streams.
  * Reconstructed anonymized frames into output videos and securely stored them in AWS S3.
  * Containerized pipeline components using Docker Compose and orchestrated them via Airflow DAGs.

Technical Skills

AI: Machine Learning, Data Analysis, Data Exploration, Data Wrangling, Data Visualization, Deep Learning, LLM, Data science, SpaCy, HuggingFace, NLTK, Transformers, GPT, OpenAI, BERT, NLP, Pandas, Numpy, Scikit-Learn, Pycaret, Pytorch, Tensorflow, Dask, Ollama, AWS Bedrock, AWS AgentCore

Gen AI:Claude, AWS Bedrock, FastAPI, MongoDB,Langchain, LangGraph, MCP, ChromaDB, Vector Database, RAG, AI Agents, PyDanticAI, Amazon Textract, OCR tools, Mircosoft Foundry, Azure OpenAI, Azure ML, Cognitive Services

DevOps: Terraform, GCP, Azure, AWS, EKS, ArgoCD, Helm, CI/CD

Versioning: Git, Github, Liquibase, DVC

Data Engineering:Temporal, Dask, Apache Airflow, PySpark

Education

Lovely Professional University,Jalandhar Jul 2017- Apr 2021
Bachelor of Technology - Computer Science Engineering *CGPA - 7.98*

Publications

An Overview on Plant Disease Detection Algorithm Using Deep Learning - 2021 2nd International Conference on Intelligent Engineering and Management (ICIEM)

Jowar Plant Leaf Disease Detection Using Deep Learning - 2021 9th International Conference on Reliability, Infocom Technologies and Optimization